

CO5-AT3- Level 3

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QUESTIONS

LEVEL 3 — HARD

Q1

Numbers



Q2

Control



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Level 3 — Hard (H) Permutations — Numbers

✓ Submitted

Write a program to print unique permutations of a given number

Sample Input:

Given Number: 143

Sample Output:

Permutations are:

134

143

314

341

413

431

Testcase: 1. 0

2. 111

3. 505

4. -143

5. -598

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Your Code

Output

```
1 import java.util.Scanner;
```

Your Code

```
1 import java.util.Scanner;
2 class permutations{
3     static void p(char[] a,int l){
4         if(l==a.length)
5             {System.out.println(new String(a));
6             return;}
7         for(int i=1;i<a.length;i++){
8             char t=a[l];
9             a[l]=a[i];
10            a[i]=t;
11            p(a,l+1);
12            t=a[l];
13            a[l]=a[i];
14            a[i]=t;
15        }
16    }
17    public static void main(String[] args){
18        Scanner s=new Scanner(System.in);
19        p(s.next().toCharArray(),0);
20    }
21 }
```

Output

```
431
413
413
431
314
```

Output

431

I

413

413

431

314

Output

314

I

341

341

314

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QUESTIONS

LEVEL 3 — HARD

Q1

Numbers



Q2

Control



2 / 2 saved

Level 3 — Hard (H) Steps — Control

✓ Submitted

Given an integer num, return the number of steps to reduce it to zero. In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it.

Example 1:

Input: num = 14

Output: 6

Explanation:

Step 1) 14 is even; divide by 2 and obtain 7.

Step 2) 7 is odd; subtract 1 and obtain 6.

Step 3) 6 is even; divide by 2 and obtain 3.

Step 4) 3 is odd; subtract 1 and obtain 2.

Step 5) 2 is even; divide by 2 and obtain 1.

Step 6) 1 is odd; subtract 1 and obtain 0.

Testcase: 1. n = 5

2. n = 10

3. n = 12

4. n = 18

5. n = 20

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Your Code

Output

Testcase: 1. n = 5

2. n = 10

3. n = 12

4. n = 18

5. n = 20

Your Code

```
1 import java.util.Scanner;
2 class steps{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int num=sc.nextInt();
6         while(num!=0){
7             if(num%2==0)
8                 System.out.println(num+" is even");
9                 num=num/2;
10                System.out.println("divide by 2 and obtain "+num);
11            if(num%2!=0)
12                System.out.println(num+" is odd");
13                num=num-1;
14                System.out.println("subtract 1 and obtain "+num);
15        }
16    }
17 }
```

Output

```
14 is even
divide by 2 and obtain 7
7 is odd
subtract 1 and obtain 6
6 is even
```

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Output

14 is even

divide by 2 and obtain 7

7 is odd

subtract 1 and obtain 6

6 is even

Output

6 is even

divide by 2 and obtain 3

3 is odd

subtract 1 and obtain 2

2 is even

divide by 2 and obtain 1

Output

2 is even

divide by 2 and obtain 1

1 is odd

subtract 1 and obtain 0